Drishti दृष्टि · "sight"

A voice-first seeing aid that tells blind and low-vision people what is ahead, in Hindi, Hinglish, English and Tamil.

THE PROBLEM

India has millions of blind and low-vision people. A cane finds what is at foot level, but not the car pulling out on the left, the friend who just walked up, or what the notice on the wall says.

Most assistive apps assume English and a good connection, and make you point, tap, wait and listen to a long description. On a busy street that is too slow to be useful.

WHO IT'S FOR

Blind and low-vision people in India who speak Hindi, Hinglish, English or Tamil and carry an everyday Android phone, including a ₹5,000 one.

The phone hangs on a lanyard with the camera facing the world. Drishti works **alongside a cane, never instead of one**. Long term, the same software runs on smart sunglasses.

THE SOLUTION

Say **"Drishti"** or hold the screen and speak. Everything else is automatic. The whole screen is one button.

What the user hears	How	Network
"Car left mein, paas"	Instant hazard alerts with direction and distance: on-device object detection and tracking, pre-recorded voice clips	Offline
"Rahul aapke saamne hai"	Names people the user saved, only after their spoken consent; strangers are only ever "unknown"	Offline
"Saamne kya hai?"	Any question about the scene: speech → Gnani Prisma → vision model → Gnani Timbre, spoken sentence by sentence	Online
Signs and notices	On-device text recognition (Latin and Devanagari), read aloud with Timbre	Mostly offline
"Tamil mein bolo"	Switch language, save or delete people, all by voice	—

WHY IT'S DIFFERENT

- **Speaks the user's language**, including mixed Hinglish, through Gnani Prisma (speech-to-text) and Timbre (text-to-speech).
- **Safety first when offline:** hazard alerts and face recognition run on the phone with no network delay.
- **Short, interruptible speech:** one speech queue where a vehicle alert cuts through any answer and stale alerts are dropped, not played late.

PRIVACY BY DESIGN

- No photos stored; saved people are kept only as face signatures on the phone.
- No transcripts, questions, answers, images or names logged, on the phone or the server.
- A name is spoken only when the match is clear; a wrong name is treated as worse than "unknown".
- A covered or dark camera is announced, so the user never assumes alerts are working when they aren't.

TEAM AND INDIVIDUAL CONTRIBUTIONS

Member	Contribution
Divyansh Parashar	Ideation, coding (Android app and server relay), video recording and editing
Nakshatra	Ideation

Built with: React Native + Expo (Android), on-device TFLite models (object detection, MobileFaceNet), ML Kit (faces, text), a FastAPI relay on Cloud Run, Gnani Prisma and Timbre, and a hosted vision model.